

Problem 10.17

A fund earns interest at a force of interest of

$$\delta(t) = 0.01t.$$

Calculate the value at the end of 10 years of \$8,000 invested at the end of 5 years.

Solution

For a variable force of interest, an amount invested at time s accumulates to time t according to

$$A(t) = A(s) \exp \left\{ \int_s^t \delta(r) dr \right\}.$$

The \$8,000 investment is made at time 5 and accumulates until time 10. Therefore,

$$A(10) = 8000 \exp \left\{ \int_5^{10} 0.01t dt \right\}.$$

Evaluating the integral,

$$\begin{aligned} \int_5^{10} 0.01t dt &= \left. \frac{0.01t^2}{2} \right|_5^{10} \\ &= 0.005 (10^2 - 5^2) \\ &= 0.005(100 - 25) \\ &= 0.375. \end{aligned}$$

Hence,

$$\begin{aligned} A(10) &= 8000e^{0.375} \\ &\approx 11639.93132. \end{aligned}$$

Therefore, the value of the fund at the end of 10 years is approximately

$$\boxed{\$11,639.93}.$$